

LAB : 2

Title : data types and arithmetic operators in cpp

1. Objectives

Tasks The objectives of this lab are to:

- Understand the basic structure of a C++ program.
- Learn how to use input () and output () operations.
- Explore common data types in C++.
- Apply arithmetic operators for calculations.
- Practice indentation and commenting for readability.
- Implement small programs including a calculator and problem-solving.

2. Theory

2.1 Basic Syntax

- Every C++ program begins with to enable input/output operations.
- The function is the entry point of execution.
- is used for output, while is used for input.
- Operators and are stream insertion and extraction operators.

2.2 Data Types

- int → integers (e.g.,)
- double → floating-point numbers (e.g.,)

- char → single characters (e.g.,)
- bool → true/false values (e.g.,)
- string → sequence of characters (e.g.,)

2.3 Arithmetic Operators

- Addition
- Subtraction
- Multiplication
- Division
- Modulus (remainder)

2.4 Best Practices

- Indentation: Indent code inside for readability.
- Comments: Use for single-line and for multi-line explanations.

3. Programs Implemented

3.1 Hello World & User Greeting

```
#include <iostream>

using namespace std;

int main () {

    cout << "Welcome to the C++ Lab!" << endl;           // output prompt

    string name;                                           // input from user

    cout << "Enter your name: ";                          // output prompt
```

```

    cin >> name;                                // input from user

    cout << "Hello, " << name << "! Let's start coding." << endl;    // output message

    return 0;                                    // tells the computer that program has ended
}

```

3.2 Simple Calculator

```

#include <iostream>

using namespace std;

int main() {                                     // main function starting pint

    int num1, num2;                             // variables declaration

    cout << "Enter two integers: ";             // outp[ut prompt

    cin >> num1 >> num2;                       // takes inputs from user


    int sum = num1 + num2;                     // finds sum

    int diff = num1 - num2;                   // finds deff

    int product = num1 * num2;                // find product

    double quotient = (double)num1 / num2;    // finds quoptiient

    int remainder = num1 % num2;              // finds reminder


    cout << "Sum: " << sum << endl;             // outputs sum

    cout << "Difference: " << diff << endl;      // outputs difference

    cout << "Product: " << product << endl;     // outputs product

    cout << "Quotient: " << quotient << endl;  // outputs quotient
}

```



```
        cout << "Remainder: " << remainder << endl;           // outputs reminder

    return 0;                                                    // tells the computer that program has ended
}
```

4.Lab Tasks

Task 1: Variable Types and Memory Size

```
#include <iostream>

using namespace std;

int main() {                                                    // main function starting pint

    int a = 10;                                                  // vsriable declaration and initialization

    double b = 20.5;                                             // variable declaration

    char c = 'A';                                                // character variable declaration

    bool d = true;                                               // boolean value declaration

    cout << "Integer: " << a << ", Size: " << sizeof(a) << " bytes" << endl;    // output
    cout << "Double: " << b << ", Size: " << sizeof (b) << " bytes" << endl;    // output
    cout << "Char: " << c << ", Size: " << sizeof(c) << " bytes" << endl;      // output
    cout << "Bool: " << d << ", Size: " << sizeof(d) << " bytes" << endl;      // output

    return 0;                                                    // tells the computer that program has ended
}
```

Task 2: Floating-Point Variable "pi"

```
#include <iostream>

using namespace std;

int main() {                                // main function starting pint
    double pi = 3.14;                        // vriable declaration and
    initialization
    cout << "Value of pi: " << pi << endl;    // output to display

    return 0;                                // tells the computer that program
    has ended
}
```

Task 3: Celsius to Fahrenheit Conversion

```
#include <iostream>

using namespace std;

int main() {                                // main function starting
    point
    double celsius, fahrenheit;              // variables declaration
    cout << "Enter temperature in Celsius: ";    // output message
    cin >> celsius;                            // takes input from
    user

    fahrenheit = (celsius * 9.0 / 5.0) + 32;    // farenhite formula
```

```
cout << "Temperature in Fahrenheit: " << fahrenheit << endl;    // output to display

return 0;                // tells the computer that program has
}
```

4. Results

- Successfully executed programs for input/output and arithmetic operations.
- Verified correct usage of data types and operators.
- Implemented calculator and temperature conversion programs.
- Displayed memory sizes of different variable types.
- Learned how indentation and comments improve readability.

5. Conclusion

This lab introduced the **fundamentals of C++ programming**, including syntax, input/output handling, data types, and arithmetic operations. By completing the tasks, students gained practical experience in writing simple programs and applying theoretical concepts. This foundation will be essential for more advanced programming labs.